# PAKISTAN – AGRICULTURAL CENSUS 2024 – METADATA REVIEW

## 1. Historical outline

Seven agricultural censuses (ACs) were conducted in the Islamic Republic of Pakistan: the first in 1960, then in 1972, 1980, 1990, 2000 and 2010. The most recent AC, to which the metadata and data presented here refer, was conducted in 2024 and integrated the Agricultural, the Livestock and the Agricultural Machinery Censuses[1].

## 2. Legal basis and organization

### *Legal framework*

The Agricultural Census Act 1958 provided the legal basis for conducting census operations. The Agriculture Census Organization (ACO) was established in 1958 as an attached department of the then Ministry of Agriculture under the legal cover of the Agriculture Census Act 1958. The Pakistan Bureau of Statistics (PBS) was established under the General Statistics (Reorganization) Act 2011, which brought the Agricultural Census Organization (ACO) under its fold.

### *Institutional framework and international collaboration*

The Agriculture Census Wing (ACW) within PBS is entrusted with the planning, execution, collection, analysis, and publication of data related to agricultural censuses and associated surveys. PBS collaborated with provincial administrations to ensure effective training of field staff, field operations, monitoring and supervision of agricultural census activities. PBS established Census Support Centres (CSCs) in each district under the supervision of the Deputy Commissioner, who served on behalf of PBS as the district focal person to convene district meetings, and also worked as Census Master Trainer (CMT), responsible for conducting training sessions for enumeration and offering technical guidance and close monitoring during census field operations. Technical committee meetings were conducted to finalize the census indicators and questionnaire design. FAO provided technical assistance in the sampling design for the AC 2024.

### *Census staff*

The AC 2024 involved 7 031 enumerators and supervisors from Provincial Governments.

## 3. Reference date and period

**Reference day**: the day of enumeration, for various variables including livestock, orchard area, fruit and non-fruit trees and stock variables.

**Reference periods**:

- crop year, for land variables, irrigation, and soil management practices, manure management, labour;
- the last 12 months, for crops, agricultural machinery, veterinary services, loans and sources of income; and
- from August 2023 to July 2024, for slaughtered animals.

## 4. Enumeration period

The AC 2024 was conducted in two distinct phases to account for varying climatic conditions across the country. The first phase was conducted in September 2024 in 24 districts identified as cold and mountainous areas. The second phase was launched in January 2025 until 10 February 2025 and covered the remaining areas. The PBS completed the village (*Mouza*) questionnaire in 2020 with some delay due to the Covid-19 pandemic. The AC enumeration had been planned for the end of 2020, but it was postponed. The village data was used for the AC sampling frame and sampling design. Based on the proposed sampled design, PBS planned to conduct the pilot census in 2021 and then implement the AC in the same year. However, the delays continued, and the AC was postponed again to 2023. Finally, it was carried out during 2024.

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[1] Specialized censuses for livestock and agriculture machinery have been carried out: to date, four livestock censuses (in 1976, 1986, 1996 and 2006), and five agricultural machinery censuses (in 1968, 1975, 1984, 1994 and 2004).

## 5. Scope of the census and definition of the statistical unit

The ***census scope*** covered agricultural activities (crop and animal production) and agricultural machinery.

The ***statistical unit*** was the agricultural farm/holding held and/or operated by the Government or by households, individually or collectively or under corporate arrangement during the time of census enumeration, normally used for crop production, rearing livestock, or having agricultural machinery,

***Community-level data***

A community survey, the *Mouza* Census, was conducted in 2020. The questionnaire was designed to collect *Mouza*-level data.

## 6. Census coverage

***Geographic coverage***

The AC 2024 covered the entire country[2].

***Cut-off threshold and other exclusions***

## 7. Methodology

***Methodological modality for conducting the census***

The AC 2024 adopted a sample-based classical approach, combining traditional field enumeration with digital data collection.

***Relation to other censuses***

The 7th Population and Housing Census (PHC) 2023 was used to build the AC frame for the household sector. The hardware and IT related requirements for the AC included 10 000 tablet devices with allied accessories that were re-used from the PHC to minimize costs.

***Frames***

The sampling frame for the AC was based on the *Mouza* Census 2020. Each *mouza* contains one or more Enumeration Blocks (EBs). In urban areas, the enumeration areas were the blocks. This frame was updated using information from the PHC 2023. The mapping of *Mouza* Census Frame 2020 was carried out and was updated using data from the PHC 2023. The sampling frame consisted of 45 753 *mouzas* and 103 972 blocks. A list of National Certainty Holdings (NCH) was constructed including big agricultural holdings and agricultural households with at least 100 acres (40.5 ha) of agricultural land, 50 cows/buffaloes, 200 goats/sheep, or 25 camels.

***Complete or/and sample enumeration methods***

The AC 2024 was conducted on a sample enumeration basis.

***Sample design***

The sample design of the AC 2024 considered different designs in each stratum. For the stratum of NCH, complete enumeration was used. In the rural area, *Mouzas* were stratified using variables related to cultivated land and livestock. In urban areas, various strata were constituted using livestock rearing and self-agriculture activities. *Mouzas* were selected at the first stage using various sampling techniques after proportional allocation in each stratum. Urban blocks were selected using simple random sampling at the first stage. From selected big *Mouzas* having two or more blocks already identified through PHC 2023, a single block was selected through a probability proportional to size method. Complete listing of all households having agricultural land, livestock, and agricultural machinery was carried out in selected *Mouzas* and/or Blocks. Those units in *Mouzas* with at least 20 acres (8.1 ha) of agricultural land, 20 cows/buffaloes, 50 goats/sheep or 20 camels were identified as

[2] Khyber Pakhtunkhwa (KPK), Punjab (including Islamabad District), Sindh, Balochistan, Azad Jammu & Kashmir (AJ&K), and Gilgit-Baltistan (GB), were covered as separate entities.

*Mouza* Certainty Holdings (MCH) and were completely enumerated. Also, households with only agricultural machinery, nomads and Gypsies were fully enumerated. In the rest of the holdings, a systematic random sampling was used. A total of 11 083 *Mouzas* and blocks were selected and allocated within strata using proportional allocation.

***Data collection methods***

The CAPI method was used for data collection in the AC 2024, although CATI and PAPI were also used in various regions to address infrastructure constraints.

***Questionnaire(s) and items covered***

Three questionnaires were used in the AC 2024. The first questionnaire (Form-1) was used for household listing in the selected *mouzas* or urban blocks for further selection. Form-2 was used to collect data in agricultural households. The third questionnaire (Form-3A) collected information regarding slaughtered animals in recognized slaughterhouses. The questionnaire covered 18 out of the 23 essential items recommended in the WCA 2020.[3]

The *Mouza* Census 2020 collected data at the *mouza*-level regarding important crops and irrigation sources, housing and sanitation, education and sport facilities, livestock, community infrastructure and services, natural resources and disasters, institutions, banks and source or credits, and industries and employment.

## 8. Use of technology

EAs maps were included in electronic devices. CAPI and CATI methods were used for data collection. Census results were disseminated online.

## 9. Data processing

Direct data capture was ensured by the CAPI and CATI methods using software developed by PBS. For data editing, the Data Cleaning Module developed by PBS was used. This app is a query-based automated data cleaning tool that replaced the traditional manual review methods, enabling faster, repeatable, and scalable error correction. AI-assisted cleaning features were used at initial scale to detect data anomalies.

## 10. Quality assurance

As part of its strategic planning, PBS launched a pilot census in July 2024 to evaluate the functionality and practicality of census technologies and processes, identify potential implementation challenges, and ensure system readiness. The AC 2024 used advanced technology, using tablets for data collection, geo-tagging of agricultural households, a real-time online dashboard for monitoring, the use of digital maps, a task management system, social media advertising campaigns and a hotline complaint system. These digital tools improved data accuracy, reduced errors, time and costs, and improved efficient processing and analysis. A call centre was established at PBS HQ to address all complaints, provide IT support and ensure data collection quality during the census process. However, the centre's main task was to handle complaints and the non-response from large agricultural holdings. PBS integrated a centralized SMS Gateway into its operational framework to facilitate rapid and reliable information exchange between the PBS HQ, regional offices, supervisors, and enumerators, enabling rapid dissemination of updates, technical instructions, operational alerts, and critical deadlines, streamlining field operations and supporting urgent decision-making. At each Census District Level, 157 dedicated Census Support Centres (CSC) were established to support and facilitate the census field operation. These CSC were responsible for managing a variety of tasks, including distributing and retrieving tablet devices, installing and configuring software applications, acting as control rooms, and handling complaints at the district level. For hard-to-reach areas and some specific areas where complete enumeration of agricultural households was required, listing and enumeration were completed together, reducing costs and non-response.

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[3] The following essential items were not covered: (i) 0105 Age of agricultural holder; (ii) 0107 Main purpose of production of the holding; (iii) 0501 Type of livestock system; (iv) 0902 Working time on the holding; and (v) 1201 Presence of aquaculture on the holding.

## 11. Data and metadata archiving

## 12. Data reconciliation

## 13. Dissemination of census results and microdata

Final results were published in xxxxx trough a Main Finding Report available at https://www.pbs.gov.pk/sites/default/files/agriculture/agri_census_2024/Agricultural_Census_Main_Findings_Report.pdf. Detailed results at the national and district level are available at https://www.pbs.gov.pk/content/detailed-results. The publications Key findings at a Glance at the national and some provinces levels are available at https://www.pbs.gov.pk/content/key-findings-glance. The lowest geographical level for which data is in public domain is district.

## 14. Data sources

**Pakistan Bureau of Statistics (PBS)**. 2025. 7th Agricultural Census. In: *PBS* [online]. Islamabad. [Cited 30 September 2025]. https://www.pbs.gov.pk/content/7th-agricultural-census

**Pakistan Bureau of Statistics (PBS)**. 2024. *7th Agricultural Census in Pakistan. Presentation to the side event of the Asian and Pacific Commission on Agricultural Statistics (APCAS), 30th Edition, Katmandu, Nepal, 19-24 May 2024.*

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